

CASEY C. GLICK, PhD

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R&D FLUIDIC SYSTEMS ENGINEER:

Multidisciplinary R&D engineer with eleven years experience in microfluidics, device fabrication, quality testing, and data analysis. Creative problem-solver in broad range of disciplines, excels in fast-paced collaborative environments, and applies theoretical background to anticipate and troubleshoot roadblocks.

Specializes in:

- Fluid and pneumatic instrumentation
- Experimental and prototype development
- Performance and reliability metrics
- Model-based performance testing and analysis
- Design of experiments and root-cause analysis
- Rapid prototyping, 3D printing, CAD, and design

WORK EXPERIENCE:

Principal Mechanical Engineer, Until Labs, South San Francisco, CA 2025

- Rebuilt cryofluid delivery system and modified control software to improve thermal stability and eliminate temperature overshoot in perfusion experiments
- Collaborated with software team to debug stack issues and integrate AI development tools for experimental automation and data analysis
- Developed theoretical models of perfusion dynamics to inform experimental priorities

R&D Engineer for Laminar at Johnson & Johnson, Santa Rosa, CA 2025

- Used SolidWorks and 3D printing to design and prototype catheter components, manufacturing jigs, and assembly fixtures
- Wrote python code for visual analysis of catheter movement for 3D deployment models
- Sourced mechanical components for traceability, compliance, and manufacturability

Senior Engineer for Illumina, Foster City, CA 2021 – 2023

- Designed, ran, and analyzed novel benchtop experiments to visualize complex fluid flows, using results to perform vendor selection and design integration for fluidic mixers
- Designed and integrated complex fluidic and pneumatic systems, including pumps, valves, and tubing, for early-stage prototypes and R&D systems.
- Built experimental, software, and model-based approaches to optimize company-wide fluidic techniques, reducing time needed for critical validation procedure by over 60%
- Developed instrumentation and design metrics across three sequencer development projects
- Collaborated locally with diverse team, and with remote groups in Singapore and London
- Partnered with biology and chemistry teams to optimize workflows and ensure system robustness

Postdoctoral Researcher for Facebook Reality Labs, Redmond, WA 2017 – 2019

- Development of fluidic control systems for next-generation haptic gloves
- Developed and experimentally validated dynamic models for fluidic valve system
- Designed and implemented new components and procedures for in-house microfluidic testbed, enabling up to 10 kHz fluidic measurements and controls
- Worked with cross-functional team on component design, simulation, and data analysis
- Organized and led team patent process, resulting in over 15 team patents

Graduate Researcher and Laboratory Manager, UC Berkeley 2011 – 2017

Principal Investigator: Professor Liwei Lin

- Research: (1) Rapid prototyping of microfluidics using 3D-printed molds, (2) Microfluidic circuitry via 3D printing, (3) Microfluidic circuitry via optofluidic lithography, (4) Analytic foundations of microfluidic circuitry; (5) 3D printed microfluidic mixing devices
- Personally mentored 6-10 undergraduates per year in hands-on laboratory research
- Developed and implemented lab safety and maintenance protocols; directed lab renovation

Graduate Student Instructor, UC Berkeley 2016

Course: BioE 124/224 *Basic Principles of Drug Delivery*, Professor Niren Murthy

Summer Undergraduate Research Fellow, HKUST, Hong Kong, China 2009

- Research: Electrorheological Fluid-Based “Universal” Microfluidic Logic Gates (*see Awards*)

EDUCATION

University of California, Berkeley

Ph.D. in Physics

Thesis: “Microfluidic Circuitry via Additive Manufacturing”

California Institute of Technology

B.S. Applied Physics, B.S. Geochemistry, minor in History

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RESEARCH/ACADEMIC SKILLS AND LANGUAGES

- Technical: Pneumatic and fluidic valves, sensors and instrumentation, 3D printing, clean room test bench development, model-based engineering, verification and validation (V&V), MEMS, mill and lathe machining, GD&T, failure modes effect analysis (FMEA), design of experiments (DOE), design for manufacturing (DFM), garment manufacture
- Software: Python, Mathematica, LaTeX, SolidWorks, CAD, Image Processing
- Academic: Fluid dynamics and microfluidics, thermal control and thermodynamics, statistics, analog and digital circuits, signal processing, mathematical methods, modeling, order-of-magnitude estimation, geochemistry, music theory and performance
- General: Systems engineering, patent preparation, Six Sigma, first responder training
- Languages: English (fluent, trained in public speaking, technical writing), Japanese (casual), Past experience with introductory Spanish

AWARDS AND COMPETITIONS

- Caltech-Intellectual Ventures Invention Competition: First Place (2010)
 - Patent: “*Electrorheological Fluid-Based ‘Universal’ Microfluidic Logic Gates*”
- McKinney Writing Competition: First Place (2010)
- World Scholar’s Cup: Second Place and Debate Champion (2007)
- California Academic Decathlon DII State Champion (2006 and 2007)

JOURNAL ARTICLES, CONFERENCE PROCEEDINGS, AND PATENTS

- [18] A. Agrawal, et al., US Patent Application №: 19/320,685;
- [17] C.C. Glick, A. Stanley, S. Keller, J. Lindsay, E. Roby, B. Manson, S. Türkylmaz, Facebook Reality Labs
US Patent №: 11,054,056; 11,098,737; 11,125,255; 11,131,331; 11,143,218;
11,231,055; 11,236,846; 11,371,619; 11,441,702; 11,480,262
- [16] A. A. Stanley, et al., “Lumped-Parameter Response Time Models for Pneumatic Circuit Dynamics.” *ASME* (2021).
- [15] E. Sweet, et al., “3D printed chaotic mixer for low Reynolds number microfluidics.” In *2019 Micro Electro Mechanical Systems (MEMS)*, IEEE, pp. 2258-2261.
- [14] R.D. Sochol, et al., “3D printed microfluidics and microelectronics.” *Microelectronic Engineering* (2018).
- [13] C. C. Glick, et al., “Rapid Assembly of Multilayer Microfluidic Structures.” In *Micro Electro Mechanical Systems (MEMS) (2018)*, IEEE, pp. 1241-1244.
- [12] C. C. Glick “Microfluidic Circuitry via Additive Manufacturing.” Dissertation, UC Berkeley (2017).
- [11] C. C. Glick, et al. “Rapid Assembly of Multilayer Microfluidic Structures via 3D Printed Transfer Molding and Bonding.” *Microsystems and Nanoengineering* (2016).
- [10] C. C. Glick et al. “Fabrication of double-sided microfluidic structures via 3D printed transfer molding.” *Hilton Head 2016 Workshop (2016)*, Transducer Research Foundation.
- [9] R. D. Sochol, et al. “3D printed microfluidic circuitry via multijet-based additive manufacturing.” *LabChip* (2016).
- [8] C. Yang, et al. “3D printed RF passive components by liquid metal filling.” In *2015 Micro Electro Mechanical Systems (MEMS)*, IEEE, pp. 261–264.
- [7] C. C. Glick et al. “Single-layer microfluidic current source via optofluidic lithography.” In *2015 Micro Electro Mechanical Systems (MEMS)*, IEEE, pp. 551–554.
- [6] C. C. Glick “Generating binary states using a microfluidic channel.” U.S. Patent №: 8,844,570 (2014).
- [5] R. D. Sochol et al. “Single-layer microfluidic ‘disc’ diodes via optofluidic lithography for ultra-low Reynolds number applications.” In *17th Int. Conf. Miniaturized Systems for Chemistry and Life Sciences (MicroTAS 2013)*.
- [4] C. C. Glick, et al. “Pressure gain in single-layer microfluidics devices via optofluidic lithography.” In *17th Conference on Solid-State Sensors, Actuators and Microsystems, Transducers & Eurosensors (2013)*, IEEE, pp. 404–407.
- [3] R. D. Sochol, et al. “Single-layer microfluidic spring diodes via optofluidic lithography for ultra-low Reynolds number applications.” In *17th Conference on Solid-State Sensors, Actuators and Microsystems, Transducers & Eurosensors XXVII (2013)*, IEEE, pp. 2201–2204.
- [2] R. D. Sochol, et al. “Single-layer ‘domino’ diodes via optofluidic lithography for ultra-low Reynolds number applications.” In *2013 Micro Electro Mechanical Systems (MEMS)*, IEEE pp. 153–156.
- [1] W. Wen et al. “Fluidic logic gates and apparatus for controlling flow of ER fluid in a channel.” International Patent WO2011115959A1 (2011).